

Conceptual content of the angle-resolved time delay (RTD) beyond the differential cross section (DCS)

Setup and exact identities

Atomic units throughout ($\hbar = 1$); SI factors are restored where they matter. Let $\mathcal{D}(E, \hat{\mathbf{k}}) = \mathcal{D}(E, \vartheta, \varphi) = R + iI = |\mathcal{D}| e^{i\Phi}$ be the one-photon dipole matrix element to the energy-normalised incoming-wave continuum state, so that Φ is fixed by the asymptotic Coulomb plus short-range scattering phases. We define the Wigner delay $\tau_W \equiv \partial_E \Phi$ (in SI, $\tau_W = \hbar \partial_E \Phi$), and the *dipole intensity*

$$\sigma(E, \hat{\mathbf{k}}) \equiv |\mathcal{D}(E, \hat{\mathbf{k}})|^2, \quad (1)$$

which equals the physical DCS up to a smooth kinematic prefactor $C(E) = 4\pi^2\omega/c$ that is independent of $\hat{\mathbf{k}}$ and cancels in every angular comparison and ratio below. With these conventions the following two decompositions of the energy derivative are *exact algebraic identities*, not approximations:

$$\partial_E \ln \mathcal{D} = \underbrace{\frac{\sigma'}{2\sigma}}_{\partial_E \ln |\mathcal{D}|} + i\tau_W, \quad (2)$$

$$\mathcal{D}^* \mathcal{D}' = \frac{1}{2}\sigma' + i\mathcal{J}, \quad \mathcal{J} \equiv \text{Im}[\mathcal{D}^* \mathcal{D}'] = RI' - R'I = \sigma \tau_W. \quad (3)$$

Both follow directly from $\mathcal{D} = |\mathcal{D}|e^{i\Phi}$: $\mathcal{D}^* \mathcal{D}' = |\mathcal{D}||\mathcal{D}'| + i|\mathcal{D}|^2\Phi'$, whose real part is $\frac{1}{2}(|\mathcal{D}|^2)' = \frac{1}{2}\sigma'$ and whose imaginary part is $|\mathcal{D}|^2\Phi' = \sigma\tau_W = \mathcal{J}$. Equation (2) isolates the delay τ_W by dividing through by σ ; (3) keeps the same information in the undivided bilinear $\mathcal{D}^* \mathcal{D}'$. DCS lives in $\sigma = |\mathcal{D}|^2$; RTD lives in τ_W ; there is no algebraic map from one to the other, so a literal numerical subtraction of DCS and RTD is meaningless (different units, different mathematical content). The substantive question — what does RTD add beyond DCS — is made precise below.

Why $\mathcal{J}$, and not the normalised delay, is the natural RTD-content variable

Three properties distinguish $\mathcal{J}$ from the normalised delay $\tau_W = \mathcal{J}/\sigma$. All three are provable from (3).

1. **Geometric content (exact).** $\mathcal{J} = \text{Im}[\mathcal{D}^* \partial_E \mathcal{D}]$ has the algebraic form of a probability current with energy in place of position. In the Argand plane of $\mathcal{D}$ it equals twice the signed rate at which the vector $\mathcal{D}(E)$ sweeps area, $\mathcal{J} = 2 dA/dE$ with $dA = \frac{1}{2}(R dI - I dR)$, so it isolates the phase-accumulating (rotational) motion of $\mathcal{D}(E)$ from the amplitude-changing (radial) motion $\frac{1}{2}\sigma'$. We call $\mathcal{J}$ the *spectral phase-current density*. We stress that it is *not* a conserved current: E is a parameter, not a dynamical coordinate, and there is no continuity equation $\partial_t \rho + \partial_E \mathcal{J} = 0$. The name refers to the algebraic structure only.
2. **Linearity under incoherent averaging (provable).** Because σ and $\mathcal{J}$ are both bilinear in $\mathcal{D}$, and $\mathcal{D}$ is linear in the field polarisation ϵ , both are linear functionals of the polarisation

density matrix and of any incoherent population distribution. For an incoherent ensemble $\{p_a, \mathcal{D}_a\}$ (e.g. unpolarised light, or randomly oriented molecules),

$$\sigma_{\text{ens}} = \sum_a p_a |\mathcal{D}_a|^2, \quad \mathcal{J}_{\text{ens}} = \sum_a p_a \text{Im}[\mathcal{D}_a^* \mathcal{D}'_a], \quad (4)$$

both sum additively. The normalised delay does not: $\tau_{W,\text{ens}} = \mathcal{J}_{\text{ens}}/\sigma_{\text{ens}}$ is a σ -weighted average of member delays, not a population average. Consequently $\mathcal{J}$ is the correct quantity to compare against any polarisation- or orientation-averaged measurement; this is illustrated by the bottom row of Fig. 1.

3. **Regularity at amplitude zeros (provable).** Where $\sigma \rightarrow 0$ — Cooper minima, multi-centre interference nodes — the normalised delay $\tau_W = \mathcal{J}/\sigma$ diverges. This divergence is a genuine feature of the phase (the phase rotates rapidly through a near-zero of $\mathcal{D}$), *not* a numerical artefact; but it does not correspond to a measurable delay, because the photoelectron yield vanishes there. By contrast $\mathcal{J}$ is yield-weighted: at a simple zero $\mathcal{D} \sim a(E - E_0)g(E)$ one finds $\mathcal{J} = \mathcal{O}((E - E_0)^2) \rightarrow 0$, so $\mathcal{J}$ is finite (and vanishing) at the zero while the delay diverges. $\mathcal{J}$ therefore reports the same phase information without the unmeasurable singularity.

Operational form of the “subtract DCS and RTD” question

The meaningful version of the PI’s “subtraction” is whether the angular pattern of $\mathcal{J}$ at fixed energy is proportional to that of σ . Define the angular inner product $\langle f, g \rangle_E = \int d\Omega f(E, \hat{\mathbf{k}}) g(E, \hat{\mathbf{k}})$ (with $d\Omega$ the solid-angle element), the best proportionality constant, and the residual fraction

$$\lambda(E) \equiv \frac{\langle \mathcal{J}, \sigma \rangle_E}{\langle \sigma, \sigma \rangle_E}, \quad \eta(E) \equiv \frac{\| \mathcal{J} - \lambda(E) \sigma \|_E}{\| \mathcal{J} \|_E}. \quad (5)$$

$\eta(E)$ is the fraction of the angular structure of $\mathcal{J}$ that is orthogonal to the angular structure of σ , computed entirely within the same calculation (no external reference is required). It has a sharp interpretation: $\eta(E) = 0$ if and only if the delay $\tau_W(E, \hat{\mathbf{k}}) = \mathcal{J}/\sigma$ is *independent of angle* at that energy, i.e. RTD carries no angular information beyond DCS; $\eta(E) \rightarrow 1$ means the RTD angular pattern is uncorrelated with the DCS angular pattern. This is a diagnostic, not a unique one — it depends on the chosen angular measure — but with the measure stated it is well defined and reproducible.

A hierarchy: when does RTD add nothing, and when does it add a lot?

The key physical fact is that $\mathcal{J}$ is generated by *final-state scattering*. This is cleanest in the plane-wave (first-Born) final-state limit, where

$$\mathcal{D}(\mathbf{k}) \propto \int e^{-i\mathbf{k} \cdot \mathbf{r}} (\boldsymbol{\epsilon} \cdot \mathbf{r}) \psi_i(\mathbf{r}) d^3r, \quad (6)$$

the Fourier transform of $(\boldsymbol{\epsilon} \cdot \mathbf{r}) \psi_i$ with *no final-state phase*.

(a) Single centre, plane-wave final state. For a real initial orbital ψ_i of definite parity, the integrand in (6) is real and of definite parity, so $\mathcal{D}(\mathbf{k})$ has a constant global phase: $\mathcal{D} \in \mathbb{R}$ (orbital odd) or $\mathcal{D} \in i\mathbb{R}$ (orbital even). In either case Φ is independent of E , hence $\tau_W = 0$, and $\mathcal{D}^* \mathcal{D}' \in \mathbb{R}$ gives $\mathcal{J} \equiv 0$ identically. RTD adds nothing.

(b) Two identical centres at $\pm \mathbf{R}/2$, plane-wave final state. A definite-parity (gerade/ungerade) two-centre orbital gives, in the Cohen–Fano LCAO form, a real structure factor

$\cos(\mathbf{k} \cdot \mathbf{R}/2)$ or $\sin(\mathbf{k} \cdot \mathbf{R}/2)$ multiplying (6). This modulates $|\mathcal{D}|$ — producing the Cohen–Fano two-centre interference fringes in σ — but introduces no energy-dependent phase, so $\mathcal{D}$ keeps its constant global phase and $\mathcal{J} \equiv 0$. *At this level the full two-centre diffraction pattern lives entirely in DCS; RTD is identically zero.* The conclusion is specific to this limit, and shows that diffraction *fringes* per se do not require RTD.

What turns $\mathcal{J}$ on. Replacing the plane wave by the true scattering state imprints energy-dependent partial-wave phases $\eta_\ell(E)$ on $\mathcal{D}$. These make Φ vary with E (so $\tau_W \neq 0$) and, through inter-channel interference, with angle — which is precisely what makes $\mathcal{J} \neq 0$. **Thus $\mathcal{J}$ is a direct, yield-weighted measure of the final-state scattering phase imprinted on the photoelectron by the potential it leaves through.** This is the “on the way out” content: absent for a free electron, present whenever the photoelectron scatters.

(c) Full cubic/octahedral C_8F_8 (this work). The anisotropic orbital and the multi-centre molecular potential mix many partial waves with E -dependent phases,

$$\mathcal{D}(E, \vartheta, \varphi) = \sum_{\ell m} c_{\ell m}(E) e^{i\eta_\ell(E)} Y_{\ell m}(\vartheta, \varphi), \quad (7)$$

so the relative phases $\eta_\ell - \eta_{\ell'}$ shape Φ in a way not tied to $|\mathcal{D}|$. Figure 1 (right column) shows the resulting $\mathcal{J}$: sign domains, zero-curves where $\text{Im}[\mathcal{D}^* \mathcal{D}']$ changes sign, and angular structure with no counterpart in σ . The residual fraction $\eta(E)$ of (5) — which is nonzero precisely when the delay τ_W is angle-dependent — is finite across the energy range, and is our quantitative answer to “what does RTD add beyond DCS?”.

Relation to attosecond measurements (remark)

The bilinear (3) is the object that appears in the finite-difference structure of two-colour attosecond observables. Expanding the two-energy correlator (an exact Taylor identity),

$$\mathcal{D}^*(E - \varepsilon/2) \mathcal{D}(E + \varepsilon/2) = |\mathcal{D}|^2 + i\varepsilon \mathcal{J}(E) + \mathcal{O}(\varepsilon^2). \quad (8)$$

In the standard soft-photon factorisation of the two-photon (RABBITT) amplitude, the continuum–continuum-corrected sideband Fourier coefficient at 2ω has, to leading order in the IR photon energy ($\varepsilon = 2\omega$), real part $\propto \sigma$ and imaginary part $\propto \varepsilon \mathcal{J}$. Hence $\mathcal{J}$ is recoverable as the quadrature (imaginary) projection of the measured complex sideband amplitude, without dividing by σ . We emphasise the limits of this statement: it holds within the factorisation approximation (it breaks near intermediate-state resonances), it is leading order in ε (the $\mathcal{O}(\varepsilon^2)$ term is not negligible at 800 nm), and it requires the same continuum–continuum phase calibration already used to extract delays. It is therefore an equivalent, better-conditioned re-parameterisation of the standard amplitude/phase analysis near nodes — *not* an independent or more fundamental measurement.

Summary

(i) RTD supplies the quadrature component of the bilinear $\mathcal{D}^* \mathcal{D}'$, complementary to the in-phase component σ . (ii) The clearest cases where it adds nothing are definite-parity orbitals with a plane-wave final state — one or two identical centres alike — where $\mathcal{J} \equiv 0$ while DCS already carries the full two-centre interference pattern. (iii) $\mathcal{J}$ is switched on by final-state scattering and measures, yield-weighted, the scattering phase the potential imprints on the outgoing electron; for C_8F_8 the anisotropic multi-centre potential makes this phase strongly angle-dependent. (iv) The appropriate “subtraction” is the orthogonal residual $\eta(E)$ of (5), which is nonzero iff the delay is angle-dependent; a literal difference of incommensurate quantities is not meaningful. (v)

Molecular-frame polar plots at $\phi = 0$ (top: fixed Euler $(\alpha, \beta) = (0^\circ, 0^\circ)$; bottom: isotropic polarisation average)
radius = E_{kin} [eV]

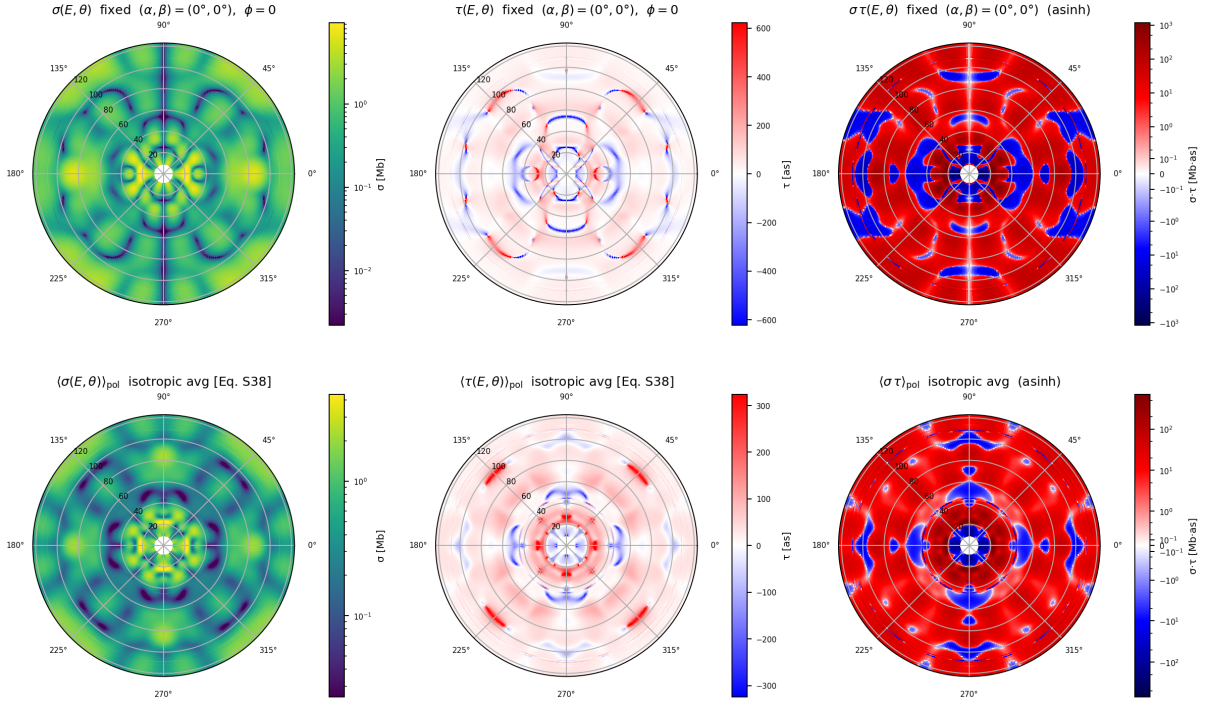


Figure 1: C_8F_8 molecular frame; radius = E_{kin} , azimuth = ϑ , at fixed φ . Top: fixed Euler orientation. Bottom: isotropic polarisation average. Left, σ ; centre, the normalised delay τ_W ; right, $\sigma\tau_W \propto \mathcal{J} = \text{Im}[\mathcal{D}^*\mathcal{D}']$ (the kinematic prefactor $C(E)$ is smooth and $\hat{\mathbf{k}}$ -independent, so it does not affect the sign or angular structure of $\mathcal{J}$). The fine angular fringes are multi-centre photoelectron diffraction; their appearance in the right column (and not only in σ) is *consistent with* beyond-Born multiple scattering and partial-wave coherence in the molecule, and is identically forbidden in the plane-wave two-centre limit of (b). The divergent stripes that decorate the nodes of σ in the central (τ_W) column are tamed in the right ($\mathcal{J}$) column, reflecting the yield weight in $\mathcal{J}$; the divergences are real phase features but carry no electron yield. The bottom-right panel illustrates that polarisation averaging acts linearly on $\mathcal{J}$ but not on τ_W .

Under incoherent (polarisation, orientation) averaging $\mathcal{J}$ is linear and τ_W is not, so $\mathcal{J}$ is the correct observable for averaged theory–experiment comparison.

We therefore introduce $\mathcal{J}(E, \hat{\mathbf{k}}) = \text{Im}[\mathcal{D}^* \partial_E \mathcal{D}]$ as a spectral phase-current density for molecular photoionisation. It is algebraically equivalent to the product of dipole intensity and Wigner delay, but as a bilinear it is regular at zeros of $\mathcal{D}$, linear under incoherent ensembles, and supplies a well-defined residual measure of the RTD content not captured by DCS alone.